

Assessment sheet 1 - to be handed in by 1700, 11th February 2026

1. This question enables us to see how the solution to a nonlinear equation can differ when we solve it subject to two different initial conditions.

- (a) Firstly, solve the following 1st order nonlinear ordinary differential equation for $y = y(x)$ given by:

$$(y - 2) \frac{dy}{dx} = 16x. \quad (1)$$

- (b) We will consider the two possible initial conditions:

$$y(0) = 1, \quad (2)$$

and

$$y(0) = 2. \quad (3)$$

By appealing to the relevant Existence and Uniqueness Theorem (i.e. without solving the equations), which of these two initial conditions is guaranteed to yield a unique solution on some interval around $x = 0$? Justify your answer.

- (c) Verify your answer to Q1b by directly solving the two initial value problems (i.e. solve (1) subject to (2) and solve (1) subject to (3)).
2. Determine whether or not it is guaranteed that a unique solution exists to the following initial value problem for $y = y(x)$, and give the interval if it is.

$$x(x^2 - 4)y'''(x) - 7xy''(x) + 9y'(x) - x^2y(x) = 0 \quad (4)$$

$$y''(x_0) = 8, \quad y'(x_0) = 10, \quad y(x_0) = -27, \quad x \in \mathbb{R} \quad (5)$$

when

- (a) $x_0 = -3$;
 (b) $x_0 = -1$.
 (c) $x_0 = 1$.

Give your reasoning.